

Processing, Training & Deployment

The Core SageMaker AI Workflow

AWS Machine Learning Associate • Part 12

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What We'll Cover



The big picture



Data processing



Training



Deployment



Running endpoints well



Automating with Pipelines

The Big Picture

1



Raw Data

Lives in Amazon S3

2



Processing Job

Cleans & transforms, writes back to S3

3



Training Job

Learns from the data, saves a model to S3

4



Endpoint

A live address that serves the model

5



Predictions

Applications get answers

S3 is the hub: each stage reads from it and hands its output to the next.

Data Processing: Processing Jobs



Input

Data in Amazon S3





Processing Job

Your script + a container, on managed instances





Output

Results written back to S3

Cleaning data

Feature engineering

Train / test split

Model evaluation

Pay only while the job runs. Ready-made containers for scikit-learn & Spark, or bring your own.

Choosing a Processing Tool



Processing Jobs

Run your own script on demand or on a schedule



Data Wrangler

Visual, point-and-click transforms with little or no code



AWS Glue

Serverless ETL for moving & transforming data across many sources



Amazon EMR (Spark)

Very large-scale data processing

All four read from and write to S3, so they fit together nicely.

Training: How a Training Job Works

1



Point to Data

Your training data in Amazon S3

2



Choose Algorithm

A container image, plus instance type & count

3



Train

SageMaker launches instances, brings in data & trains

4



Save the Model

model.tar.gz goes to S3, then instances shut down

Logs & metrics go to CloudWatch, so you can see how training went.

Training: Your Options



Algorithm Choices

Built-in (like XGBoost), your own script with PyTorch or TensorFlow, or your own container



Input Modes

File copies data first, FastFile streams as needed, Pipe streams straight into the algorithm



Distributed Training

Spread big models or datasets across many instances



Spot & Tuning

Managed spot training cuts cost; automatic tuning improves accuracy

Deployment: Model, Config, Endpoint

1



Model

Your model artifact from S3 plus an inference container

2



Endpoint Configuration

Instance type, instance count & variants

3



Endpoint

The live HTTPS address your applications call

Created in this order. SageMaker manages the servers behind it.

Deployment: Running Endpoints Well



Auto Scaling

Adds or removes instances as traffic goes up and down



Production Variants

Send a share of traffic to a new model version for A/B testing



Multi-Model Endpoints

Host many rarely used models behind one endpoint to save money

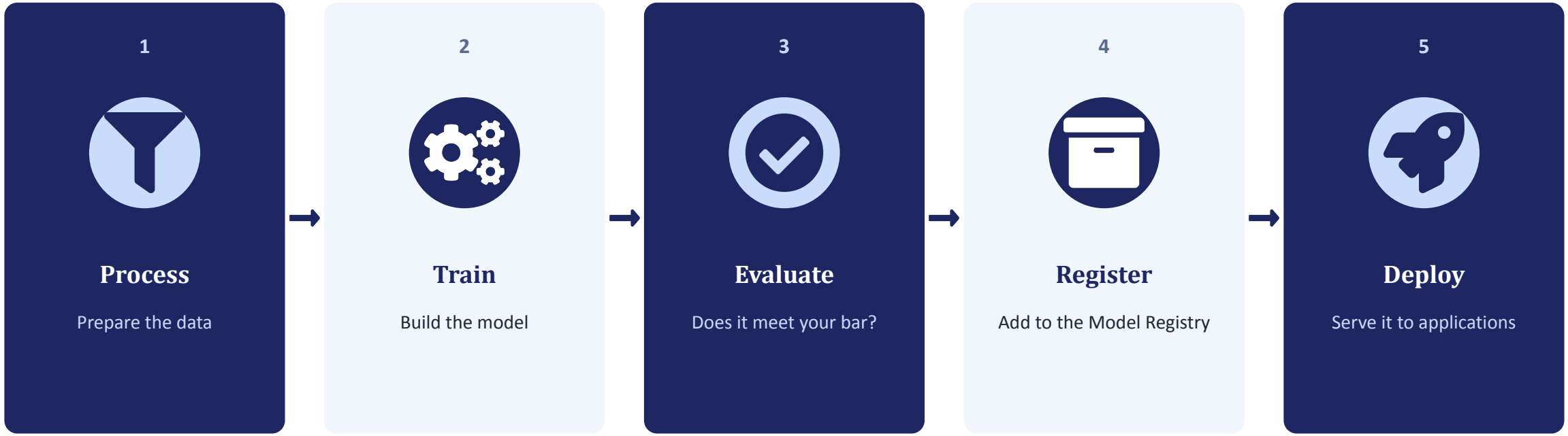


Safe Rollouts

Blue/green & canary deployments, with rollback if something goes wrong

Not real-time? Serverless, asynchronous & batch inference are options too.

Automating with SageMaker Pipelines



Only deploy if the evaluation passes. Every run is tracked and repeatable.

Recap

- Processing jobs prepare data; training jobs turn it into a model.
- Endpoints serve that model to your applications.
- S3 connects every stage, and Pipelines automate the whole flow.

Thanks for watching!